O_16_observe_0421_10R1e3

April 23, 2026

1 Observations

- 20260312: all conditions are based on fast learning (1e-3) so that we can observe more convergence across runs.
- 20260315: increased EWC strength from 1.0 to 10000.0, making EWC penalty comparable with task loss. Fixed some bugs in EWC implementation.
- 20260317: shrank down hidden dimension to 2, hopefully it will make EWC less advantageous. Originally we had dim = 4, and with this setting we got EWC > NFZ for both L1 and L2. This seems rather weird. Therefore, I doubted whether it could be due to over-powered hidden representation, thus making it best not to move any L1 parameters but to start with new dimensions, and EWC was exactly doing this, making it better than NFZ. In this version, we make hidden dimension = 2. With more limited hidden dimensions, we anticipate to see some difference.
- 20260319: Because we still saw quite obvious advantage of EWC for both Para and Vert, I doubted whether it is due to the model in general being too powerful. Therefore, I ran a simplistic FC Recon model and this is the result.
- 20260417: Additionally added WeakEWC (weight 10000 -> 1000); learning rate 1e-3 -> 1e-4.
- 20260421: Changed Parallel to tsh-tch instead of sh-ch; running different EWC weights.

Comparison is between NFZ and EWC, PARA and VERT.

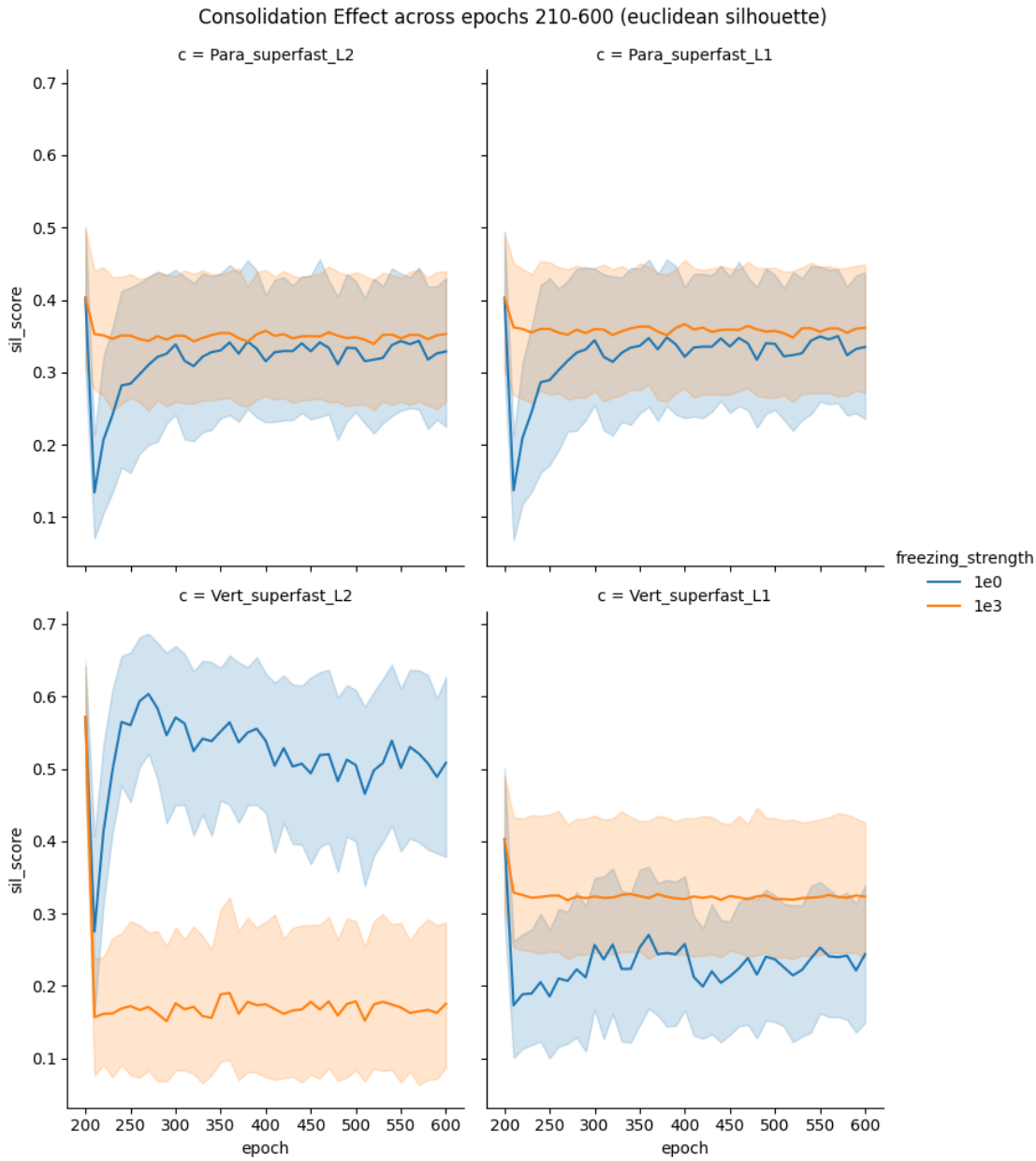
L1: s-c

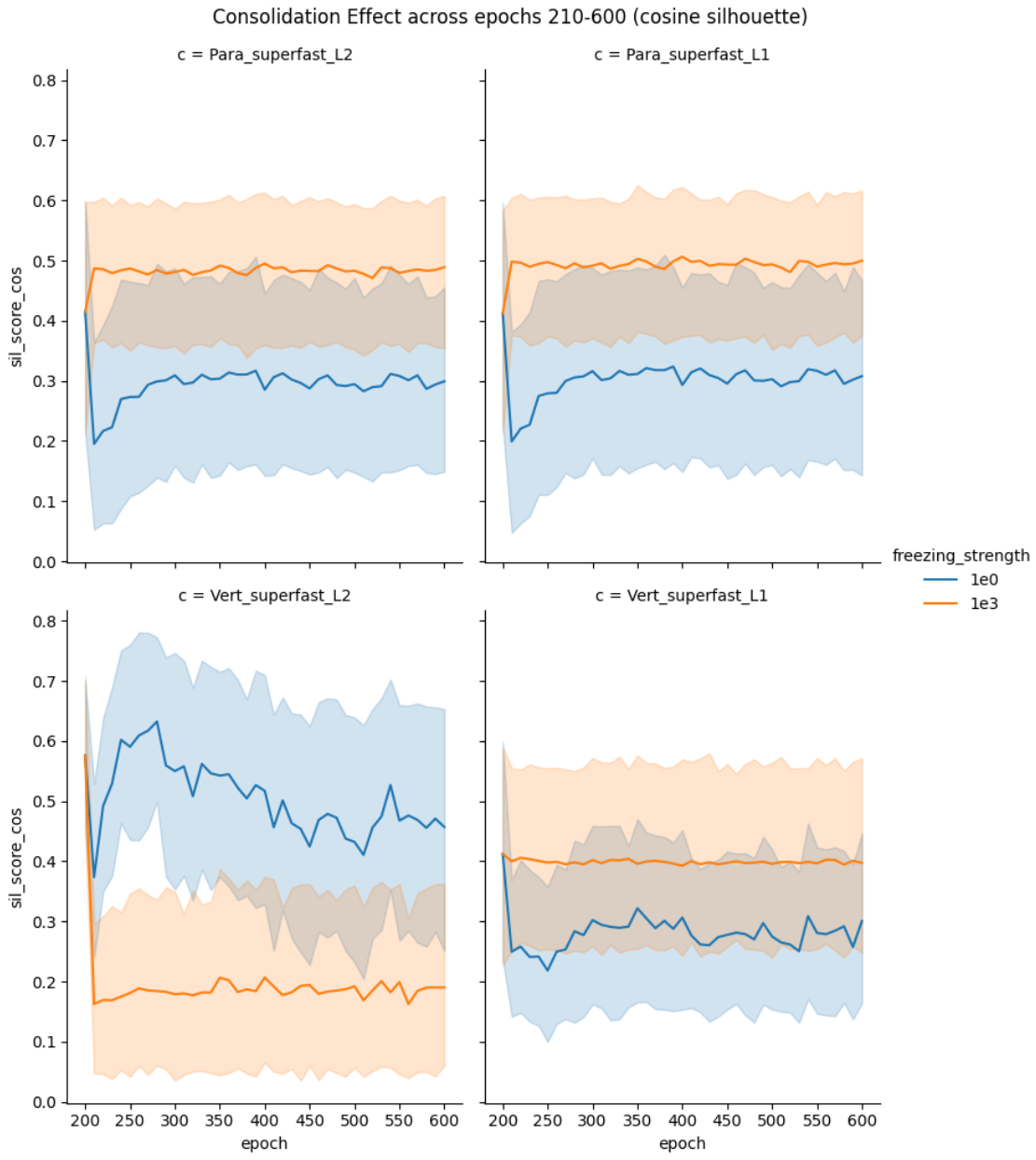
L2: tsh-tch / tsh-sh

```
scp -r franklhtan@10.64.109.137:"/nfs/franklhtan/projects/LearnCsld2025/observe_outputs/0421_*"
/mnt/data/Projects/LearnCsldLab/observations/
```

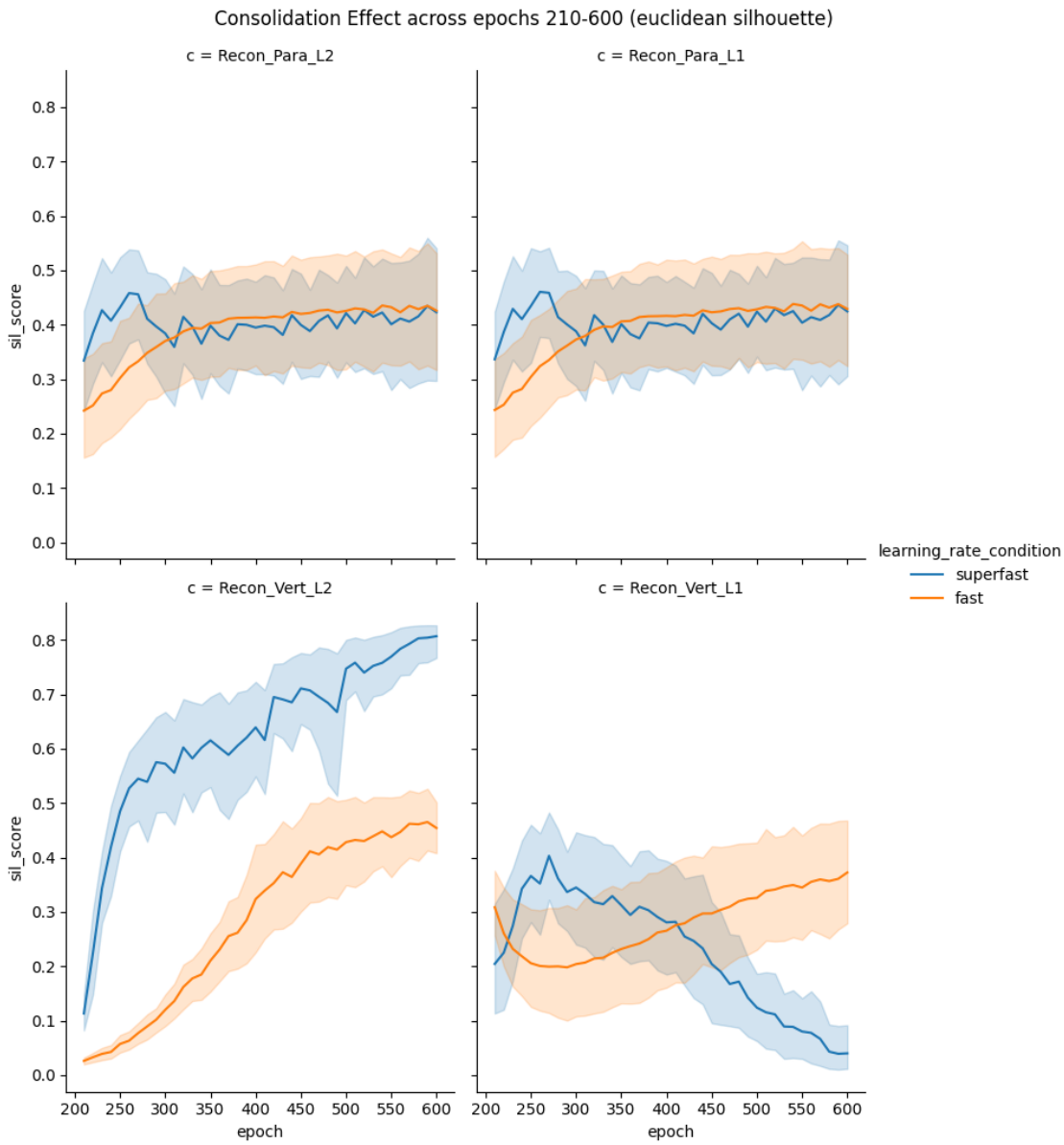
Use this code to transfer from server to lab computer.

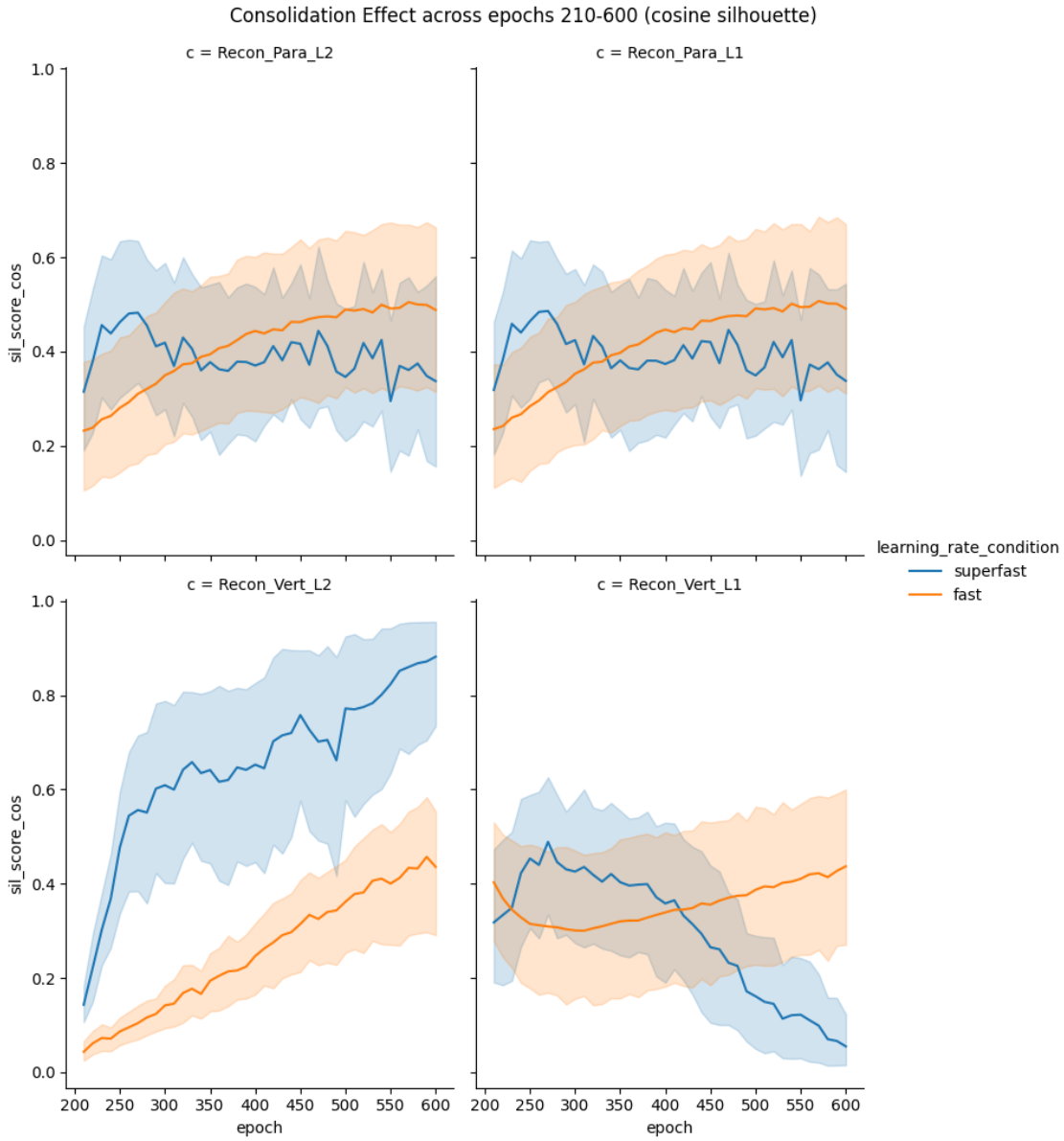
1.1 Confirm: EWC strengths



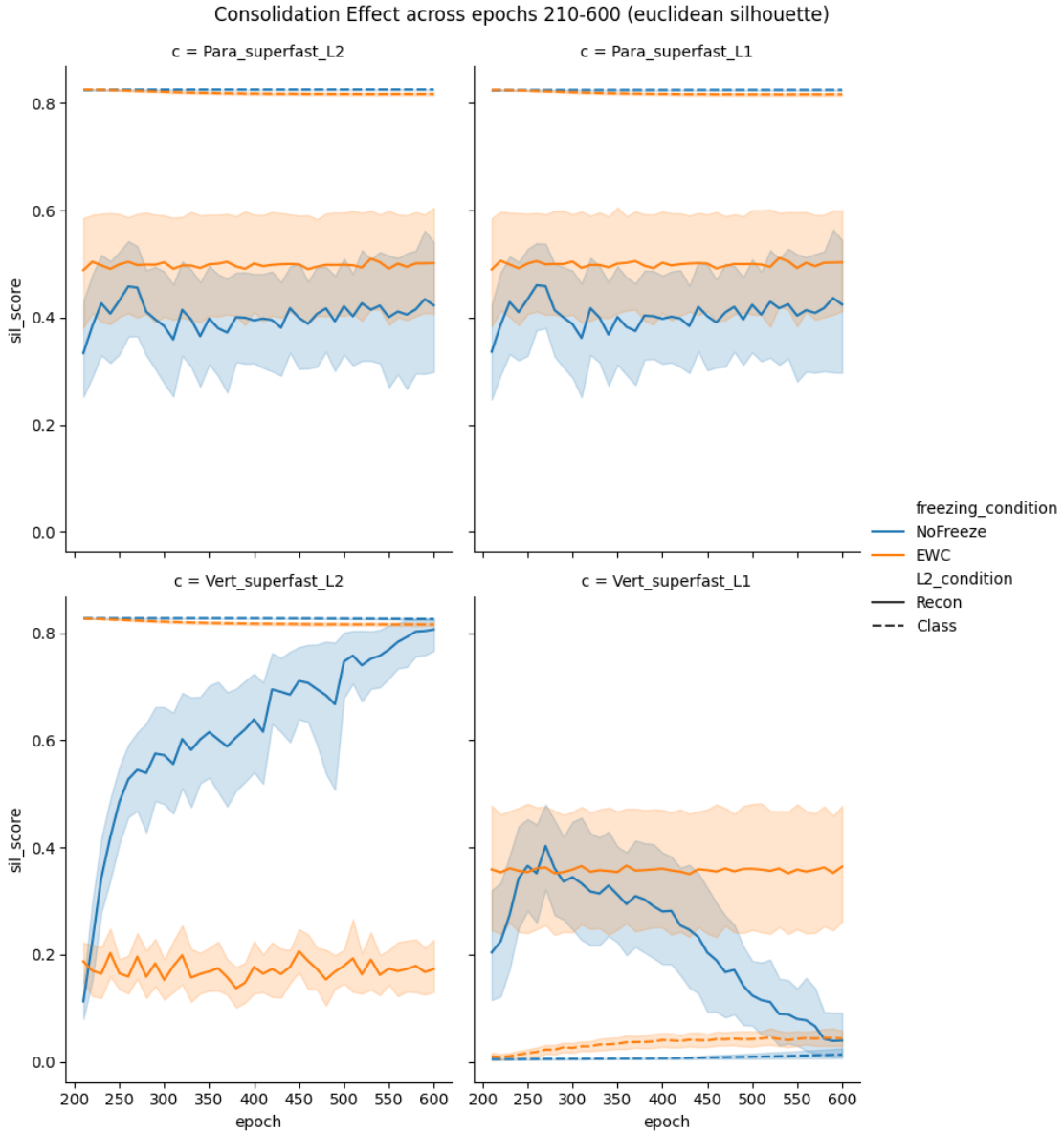

Consolidation Effect across epochs 210-600 (cosine silhouette)
c = Para_superfast_L2
c = Para_superfast_L1
c = Vert_superfast_L2
c = Vert_superfast_L1
sil_score_cos
epoch
freezing_strength
1e0
1e3

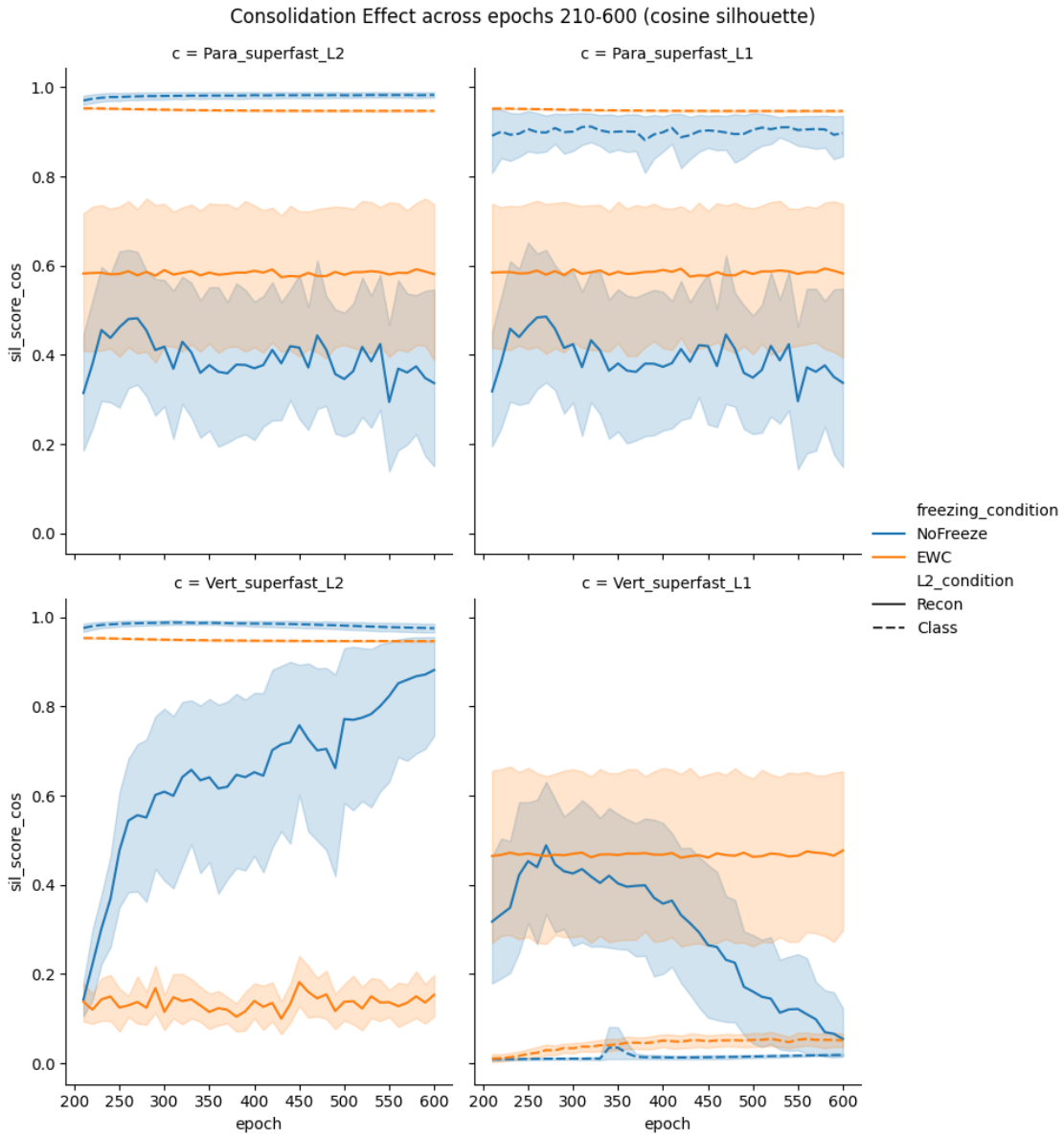
1.2 Learning Rate: 1e3 vs 1e4




Consolidation Effect across epochs 210-600 (cosine silhouette)
c = Recon_Para_L2
c = Recon_Para_L1
c = Recon_Vert_L2
c = Recon_Vert_L1
sil_score_cos
epoch
learning_rate_condition
superfast
fast

1.3 NFZ:EWC ~ RC:CL




Consolidation Effect across epochs 210-600 (cosine silhouette)
c = Para_superfast_L2
c = Para_superfast_L1
c = Vert_superfast_L2
c = Vert_superfast_L1
sil_score_cos
epoch
freezing_condition
NoFreeze
EWC
L2_condition
Recon
Class